

Yang Zhou

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RESEARCH INTERESTS

Machine learning systems, networked systems

EDUCATION

Harvard University, Cambridge, MA, USA

Ph.D. in Computer Science

June 2024

M.S. in Computer Science

November 2021

Thesis: Network-Application Co-design for Efficient Datacenters

Advisors: Minlan Yu and James Mickens

Peking University, Beijing, China

B.S. in Computer Science

July 2018

Thesis: Towards Faster and More Accurate Data Stream Processing

Advisor: Tong Yang

EMPLOYMENT

xAI, Member of Technical Staff, Pretraining

Oct 2025 -

University of California, Davis, Assistant Professor of Computer Science

July 2025 -

University of California, Berkeley, Postdoctoral researcher in Sky Computing Lab

July 2024 - now

Project: UCCL: Building Efficient GPU Communication Systems

Supervisor: Ion Stoica

Google SRG and NetInfra, Research Intern

June 2021 - May 2023

VMware Research, Research Intern

July 2020 - September 2020

Meta/Facebook, Research Collaborator

November 2019 - May 2020

SenseTime, Software Engineering Intern

March 2018 - May 2018

PROFESSIONAL SERVICE

Leadership:

- Co-Chair: ACM SIGCOMM Artifact Evaluation 2024

Program Committees:

- ACM APSys: 2026
- IEEE ICNP: 2026
- USENIX NSDI: 2026
- USENIX OSDI: 2025
- ACM EuroSys: 2026
- ACM ASPLOS: 2026
- ACM SIGCOMM Workshop on eBPF and Kernel Extensions 2024, 2025

- ACM SIGCOMM Workshop on Networks for AI Computing 2025, 2026
- ACM SIGCOMM Poster/Demo 2023, 2024, 2025
- IEEE INFOCOM: Workshop on Networking Algorithms 2020

OPEN SOURCE SOFTWARE

- UCCL, ultra and unified GPU communication library, 1300+ stars
<https://github.com/uccl-project/uccl>
- LEANN, the smallest vector index in the world for RAG, 10,000+ stars
<https://github.com/yichuan-w/LEANN>
- NEO, an LLM inference engine built to save the GPU memory by CPU offloading
<https://github.com/NEO-MLSys25/NEO>
- eTran, implementing reliable network transports in the Linux kernel with eBPF
<https://github.com/eTran-NSDI25/eTran>
- DINT, running distributed transactions in the Linux kernel with eBPF
<https://github.com/DINT-NSDI24/DINT>
- Electrode, running Paxos consensus in the Linux kernel with eBPF
<https://github.com/Electrode-NSDI23/Electrode>

PUBLICATIONS

Preprints

- [1] Xingyu Xiang, Raj Joshi, Yuhao Liu, Jiayi Yao, Chenxingyu Zhao, Junchen Jiang, **Yang Zhou**, Eddie Kohler, Minlan Yu.
ShadowServe: Interference-Free KV Cache Fetching for Distributed Prefix Caching. [\[link\]](#)
Arxiv Sep 2025
- [2] Jiarong Xing, Yifan Qiao, Simon Mo, Xingqi Cui, Gur-Eyal Sela, **Yang Zhou**, Joseph Gonzalez, Ion Stoica.
Towards Efficient and Practical GPU Multitasking in the Era of LLM. [\[link\]](#)
Arxiv Aug 2025
- [3] Shiyi Cao, Yichuan Wang, Ziming Mao, Pin-Lun Hsu, Liangsheng Yin, Tian Xia, Dacheng Li, Shu Liu, Yineng Zhang, **Yang Zhou**, Ying Sheng, Joseph Gonzalez, Ion Stoica.
Locality-Aware Fair Scheduling in LLM Serving. [\[link\]](#)
Arxiv Jan 2025
- [4] Yifan Qiao, Shu Anzai, Shan Yu, Haoran Ma, Shuo Yang, Yang Wang, Miryung Kim, Yongji Wu, **Yang Zhou**, Jiarong Xing, Joseph Gonzalez, Ion Stoica, Harry Xu.
ConServe: Harvesting GPUs for Low-Latency and High-Throughput Large Language Model Serving. [\[link\]](#)
Arxiv Oct 2024
- [5] Shuo Yang, Ying Sheng, Yilong Zhao, Joseph Gonzalez, **Yang Zhou**, Ion Stoica, Lianmin Zheng.
Post-Training Sparse Attention with Double Sparsity. [\[link\]](#)
Arxiv Aug 2024

Conference

- [6] **Yang Zhou**, Zhongjie Chen, Ziming Mao, ChonLam Lao, Shuo Yang, Pravein Govindan Kannan, Jiaqi Gao, Yilong Zhao, Yongji Wu, Kaichao You, Fengyuan Ren, Zhiying Xu, Costin Raiciu, Ion Stoica.

- UCCL-Tran: An Extensible Software Transport Layer for Machine Learning Workloads. [\[link\]](#)
OSDI 2026
- [7] Ziming Mao, Yihan Zhang, Chihan Cui, Zhen Huang, Kaichao You, Zhongjie Chen, Zhiying Xu, Zhenyu Gu, Scott Shenker, Costin Raiciu, **Yang Zhou**, Ion Stoica.
UCCL-EP: Portable Expert-Parallel Communication. [\[link\]](#)
OSDI 2026
- [8] Yilong Zhao, Xiaonan Nie, Kan Zhu, Shuang Ma, Zhichao Lai, Hongxiang Hao, **Yang Zhou**, Baris Kasikci, Ion Stoica
Unleashing Scalable Context Parallelism for Foundation Models Pre-Training via FCP
MLSys 2026.
- [9] Yichuan Wang, Shu Liu, Zhifei Li, Yongji Wu, Ziming Mao, Yilong Zhao, Xiao Yan, Zhiying Xu, **Yang Zhou**, Ion Stoica, Sewon Min, Matei Zaharia, Joseph E. Gonzalez.
LEANN: a Low-Storage Vector Index. [\[link\]](#)
MLSys 2026.
- [10] Yilong Zhao, Shuo Yang, Kan Zhu, Lianmin Zheng, Baris Kasikci, Yifan Qiao, **Yang Zhou**, Jiarong Xing, Ion Stoica.
BlendServe: Optimizing Offline Inference for Auto-regressive Large Models with Resource-aware Batch-ing. [\[link\]](#)
ASPLOS 2026
- [11] Yifan Zhang, Jianchang Su, Zixu Shen, **Yang Zhou**, Wei Zhang.
Toward Interference-Aware Scheduling for Serverless Functions via eBPF and Meta-Learning. [\[Link\]](#)
PACMI 2025. Workshop on Practical Adoption Challenges of ML for Systems.
- [12] Xiangfeng Zhu, **Yang Zhou**, Yuyao Wang, Xiangyu Gao, Arvind Krishnamurthy, Sam Kumar, Ratul Mahajan, Danyang Zhuo.
Rethinking RPC Communication for Microservices-based Applications. [\[Link\]](#)
HotOS 2025.
- [13] Xuanlin Jiang, **Yang Zhou**, Shiyi Cao, Ion Stoica, Minlan Yu.
NEO: Saving GPU Memory Crisis with CPU Offloading for Online LLM Inference. [\[link\]](#)
MLSys 2025.
- [14] Zhongjie Chen, Qingkai Meng, ChonLam Lao, Yifan Liu, Fengyuan Ren, Minlan Yu, **Yang Zhou**.
eTran: Extensible Kernel Transport with eBPF. [\[link\]](#)
USENIX NSDI 2025.
- [15] **Yang Zhou**, Mark Wilkening, James Mickens, and Minlan Yu.
SmartNIC Security Isolation in the Cloud with S-NIC. [\[link\]](#)
ACM EuroSys 2024.
- [16] **Yang Zhou**, Xingyu Xiang, Matthew Kiley, Sowmya Dharanipragada, and Minlan Yu.
DINT: Fast In-Kernel Distributed Transactions with eBPF. [\[link\]](#)
USENIX NSDI 2024.
- [17] **Yang Zhou**, Zezhou Wang, Sowmya Dharanipragada, and Minlan Yu.
Electrode: Accelerating Distributed Protocols with eBPF. [\[link\]](#)
USENIX NSDI 2023.
- [18] **Yang Zhou**, Hassan Wassel, Sihang Liu, Jiaqi Gao, James Mickens, Minlan Yu, Chris Kennelly, Paul Turner, David Culler, Hank Levy, and Amin Vahdat.
Carbink: Fault-Tolerant Far Memory. [\[link\]](#)

USENIX OSDI 2022.

- [19] **Yang Zhou**, Ying Zhang, Minlan Yu, Guangyu Wang, Dexter Cao, Eric Sung, and Starsky Wong. Evolvable Network Telemetry at Facebook. [\[link\]](#)
USENIX NSDI 2022.
- [20] **Yang Zhou**, Tong Yang, Jie Jiang, Bin Cui, Minlan Yu, Xiaoming Li, and Steve Uhlig. Cold Filter: A Meta-Framework for Faster and More Accurate Stream. Processing [\[link\]](#)
ACM SIGMOD 2018.
- [21] Tong Yang, Jie Jiang, Peng Liu, Qun Huang, Junzhi Gong, **Yang Zhou**, Rui Miao, Xiaoming Li, and Steve Uhlig. Elastic Sketch: Adaptive and Fast Network-Wide Measurements. [\[link\]](#)
ACM SIGCOMM 2018.
- [22] Omid Alipourfard, Masoud Moshref, **Yang Zhou**, Tong Yang, and Minlan Yu. A Comparison of Performance and Accuracy of Measurement Algorithms in Software. [\[link\]](#)
ACM Symposium on SDN Research (SOSR) 2018.
- [23] Xiangyang Gou, Chenxingyu Zhao, Tong Yang, Lei Zou, **Yang Zhou**, Yibo Yan, Xiaoming Li, and Bin Cui. Single Hash: Use One Hash Function to Build Faster Hash Based Data Structures. [\[link\]](#)
IEEE International Conference on Big Data and Smart Computing (BigComp) 2018.
- [24] Tong Yang, **Yang Zhou**, Hao Jin, Shigang Chen, and Xiaoming Li. Pyramid Sketch: A Sketch Framework for Frequency Estimation of Data Streams. [\[link\]](#)
VLDB 2017.
- [25] **Yang Zhou**, Peng Liu, Hao Jin, Tong Yang, Shoujiang Dang, and Xiaoming Li. One Memory Access Sketch: A More Accurate and Faster Sketch for Per-Flow Measurement. [\[link\]](#)
IEEE Global Communications Conference (Globecom) 2017.
- [26] Junzhi Gong, Tong Yang, **Yang Zhou**, Dongsheng Yang, Shigang Chen, Bin Cui, and Xiaoming Li. ABC: A Practicable Sketch Framework for Non-Uniform Multisets. [\[link\]](#)
IEEE International Conference on Big Data (BigData) 2017.

Workshop and Demo

- [27] **Yang Zhou**, Hao Jin, Peng Liu, Haowei Zhang, Tong Yang, and Xiaoming Li. Accurate Per-Flow Measurement with Bloom Sketch. [\[link\]](#)
IEEE International Conference on Computer Communications Workshops (INFOCOM WKSHPS) 2018.

Journal

- [28] Zhuochen Fan, Gang Wen, Zhipeng Huang, **Yang Zhou**, Qiaobin Fu, Tong Yang, Alex X Liu, and Bin Cui. On the Evolutionary of Bloom Filter False Positives - An Information Theoretical Approach to Optimizing Bloom Filter Parameters. [\[link\]](#)
IEEE Transactions on Knowledge & Data Engineering 2022.
- [29] Yuanpeng Li, Xiang Yu, Yilong Yang, **Yang Zhou**, Tong Yang, Zhuo Ma, and Shigang Chen. Pyramid Family: Generic Frameworks for Accurate and Fast Flow Size Measurement. [\[link\]](#)
IEEE/ACM Transactions on Networking 2021.
- [30] Tong Yang, Jie Jiang, **Yang Zhou**, Long He, Jinyang Li, Bin Cui, Steve Uhlig, and Xiaoming Li. Fast and Accurate Stream Processing by Filtering the Cold. [\[link\]](#)
The VLDB Journal 2019.

- [31] Tong Yang, Jie Jiang, Peng Liu, Qun Huang, Junzhi Gong, **Yang Zhou**, Rui Miao, Xiaoming Li, and Steve Uhlig.
Adaptive Measurements Using One Elastic Sketch. [\[link\]](#)
IEEE/ACM Transactions on Networking 2019.
- [32] **Yang Zhou**, Omid Alipourfard, Minlan Yu, and Tong Yang.
Accelerating Network Measurement in Software. [\[link\]](#)
ACM SIGCOMM Computer Communication Review 2018.

TALKS

- UCCL: Building Efficient GPU Communication Systems
UC Berkeley SkyLab Winter Retreat *January 2026*
UIUC *Dec 2025*
Northeastern, UW-Madison *Nov 2025*
Google network summit, Google NetInfra *Oct 2025*
Meta, ByteDance (Seed), SJTU IPADS *July 2025*
ByteDance (Networking), NVIDIA *June 2025*
UC Berkeley SkyLab Summer Retreat, Broadcom *May 2025*
UC Berkeley SkyLab Winter Retreat *January 2025*
- Network-Application Co-design for Efficient Datacenters
University of Toronto *April 2024*
NYU, Brown, UC Irvine, UWaterloo, UC Davis, Boston University *March 2024*
UC Santa Cruz, University of Virginia, Purdue *February 2024*
- Electrode: Accelerating Distributed Protocols with eBPF
Duke University, ACE Center for Evolvable Computing, Google, USENIX NSDI *April 2023*
Columbia University *March 2023*
- Carbink: Fault-Tolerant Far Memory
Cornell University *November 2023*
WORDS workshop *November 2022*
Microsoft Research Redmond, USENIX OSDI *July 2022*
Google *March & June 2022*
- Evolvable Network Telemetry at Facebook
USENIX NSDI *April 2022*
Boston University, Meta *March 2022*
- Cold Filter: A Meta-Framework for Faster and More Accurate Stream Processing
Harvard University *October 2018*

STUDENTS

Current Students

- Shuang Ma (PhD) *2025 - now*
- Yihan Zhang (PhD) *2025 - now*

Mentored Students

- Ziming Mao, UC Berkeley PhD *2024 - now*
UCCL (2× OSDI 2026).

- Zhongjie Chen, Tsinghua University PhD → MSRA senior researcher 2024 - now
Extensible kernel transport (NSDI 2025), (2× OSDI 2026).
- Xuanlin Jiang, Peking University undergraduate → Harvard PhD 2024
CPU offloading for online LLM inference (MLSys 2025).
- Matt Kiley, Harvard College undergraduate → Clockwork Systems 2023
Accelerating distributed transactions using eBPF (NSDI 2024).
- Yunxi Shen, Tsinghua University undergraduate → Cornell PhD 2023
Resource-efficient job scheduling in data centers.
- Xingyu Xiang, Peking University undergraduate → Harvard PhD 2023
Accelerating distributed transactions using eBPF (NSDI 2024).
- Zezhou Wang, Peking University undergraduate → University of Washington PhD 2022
Accelerating Paxos using eBPF (NSDI 2023).

GRANTS

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- AWS Research Credits on GPUs, \$477,000 2025

TEACHING EXPERIENCE

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- **Instructor** ECS 289D: Seminar in Operating Systems: Datacenter Systems for LLMs, UC Davis *Fall 2025*
 - **Guest Lecture** on UCCL, CS538: Advanced Computer Networks, UIUC *Dec 2025*
 - **Guest Lecture** on UCCL, CS7670: Large Language Model Systems (Seminar), Northeastern *Nov 2025*
 - **Guest Lecture** on far memory, CS294-252: Architectures and Systems for Warehouse-Scale Computers, UC Berkeley *Nov 2023*
 - **Teaching Assistant** for Prof. Minlan Yu, CS145: Networking at Scale, Harvard University *Spring 2021*
 - **Teaching Assistant** for Prof. Tong Yang, Algorithm Design and Analysis, Peking University *Fall 2018*

PATENTS

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- **Yang Zhou**, Hassan Wassel, Minlan Yu, Hank Levy, David Culler, and Amin Vahdat. “Fault Tolerant Disaggregated Memory”. Pending (US20230185666A1), December 2022.

AWARDS AND HONORS

-
- Google Ph.D. Fellowship in Systems and Networking 2022
 - Finalist, Meta Ph.D. Fellowship in Networking 2022
 - Graduate Fellowship, Harvard University 2018
 - Excellent Bachelor Thesis (10/327), School of EECS, Peking University 2018
 - New Academic Star Award (1/193), School of EECS, Peking University 2018
 - Arawana Scholarship (2/193), Peking University 2017
 - Pinyou Hudong Scholarship, School of EECS, Peking University 2016
 - May Fourth Scholarship, Peking University 2015